

Evaluating Fluorescence Sensitivity on Flow Cytometers: An Overview

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The current paradigms for assessing fluorescence sensitivity on flow cytometers do not provide an adequate assessment of an instrument's ability to detect and measure weak fluorescence on stained particles. The capability to resolve dimly stained populations depends on two factors: the background noise (B), and the efficiency (Q) with which the fluorescence from the fluorochrome molecules are converted to photoelectrons. Any single statistical measure of fluorescence histogram distributions will be unable to uniquely characterize an instrument. Therefore, neither of the routinely used meth-

ods (detection threshold and delta channel) measure sensitivity completely and unambiguously. We show the limitations of these methods and propose that instrument sensitivity be characterized in terms of both background noise and detection efficiency in order to determine better the capability to detect and resolve weakly fluorescent particles. Cytometry 33: 256–259, 1998. © 1998 Wiley-Liss, Inc.

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Quantitative fluorescence cytometry (QFCM) requires the proper measurement of weak photonic signals. Such measurements are subject to certain limits of sensitivity, both theoretical and practical. Despite an intuitive sense of what fluorescence sensitivity means, there is no general agreement on an actual definition of this fundamental parameter. This overview describes the basic considerations involved in defining and evaluating fluorescence sensitivity. More detailed treatments of this topic are presented in the two following reports (Wood, Chase and Hoffman, this issue).

DUAL DEFINITIONS OF SENSITIVITY: THRESHOLD AND RESOLUTION

Fluorescence sensitivity has two different definitions: threshold and resolution (3). The fluorescence threshold is the lowest signal from stained particles which a cytometer can distinguish from background (5–7,9). The concept of threshold is similar to the definition of sensitivity proposed by Steen (8) based on certain physical and electronic contexts. It requires a statistically significant difference between the mean fluorescence intensity (FI) of background signals and the mean FI of particles emitting at the fluorescence threshold level, even though the two populations may overlap almost entirely.

Resolution, the second aspect of sensitivity, is the degree to which the flow cytometer can distinguish unstained and dimly stained populations in a mixture (1,4,7). Resolution allows the identification of distinct

phenotypes. The coefficient of variation (CV) of fluorescence signals is critical for resolution, since increased CVs produce overlapping distributions.

BASIC THEORY OF WEAK PHOTONIC SIGNAL DETECTION

The fluorescence emission of stained cells is a weak photonic signal, like the signals detected by radio receivers and scintillation counters. Technologies to detect such signals share two general requirements. First, enough signal photons must be collected by the detector to differentiate them from interfering noise photons and to achieve the required photon counting precision. Second, each detected photon must be efficiently converted into a measurable electronic signal. Sensitivity is a multivariate property of the detection system; how well a detection system collects and converts FI signals determines both threshold and resolution.

Basic theory dictates certain inherent limits to the sensitivity of weak photonic signal detection. In particular, no amount of signal amplification can compensate for a flow cytometer that collects an inadequate number of photons. Photon collection itself is constrained by the indices of refraction of the various components of the

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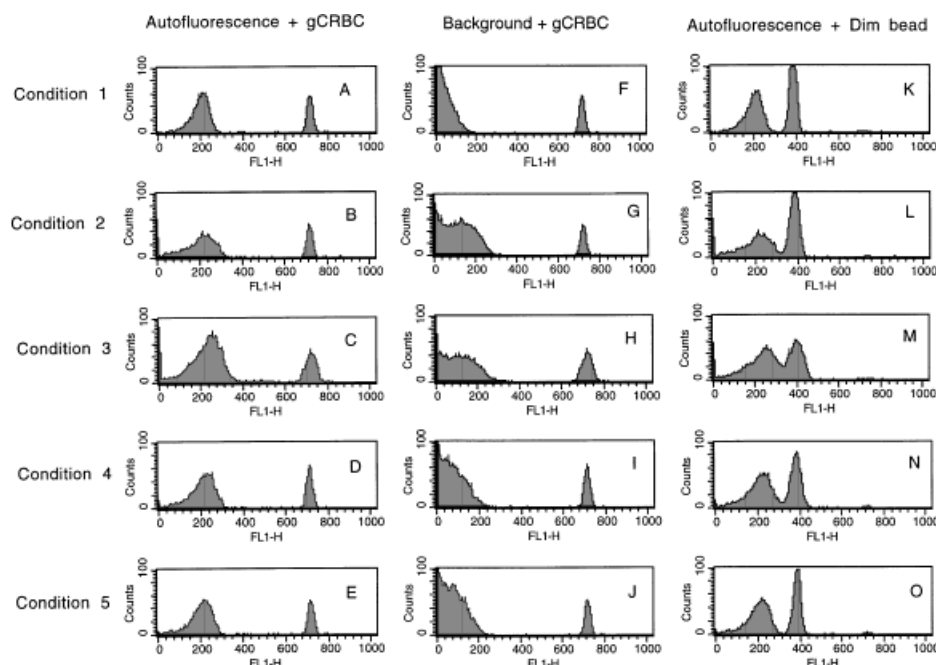


FIG. 1. Comparison of three methods for estimating green fluorescence (FL1) fluorescence sensitivity. Histograms are four-decade log with 256 channels per decade. Each column of histograms represents a different measure of fluorescence sensitivity. **A–E:** Histograms of autofluorescent unstained CaliBRITE® beads (low peak) and glutaraldehyde-fixed chicken red blood cells (gCRBC) (high peak) illustrating the delta channel method.

F–J: Histograms of background noise from a test pulse trigger (low peak) and gCRBC (high peak) illustrating the detection threshold method. **K–O:** Histograms of autofluorescent unstained CaliBRITE® beads and dim Rainbow beads illustrating the qualitative resolution method. Each row of histograms shows results for a different condition or perturbation on the optical system of the instrument. Reprinted from (3).

optical system (lenses, air, suspending liquid, flow cell, etc.) and by the phenomenon of optical sectioning, all of which limit the angle through which photons can be collected. Reflectors may be used to capture more photons, although this approach has its own drawbacks. More photons may also be collected by lengthening the collection interval (as in a scintillation counter, where the ability to detect and resolve weak signals is improved by increasing the counting time over which the photons are collected). The same concept is possible for special applications in a flow cytometer by moving the particles very slowly through the interrogation region (2).

In addition to optimizing the collection of photons, sensitivity may be improved by reducing the number of interfering noise photons. This can be accomplished with a combination of optical filters (which limit the range of photon wavelengths collected) and spatial filters (which limit the volume from which light is collected). Unfortunately, these filters also tend to reduce the number of signal photons collected, so the loss of signal photons must be balanced against the reduction in interfering photons.

The light-collection system directs the photons to a detector that converts them to a measurable electronic signal. The collection efficiency and gain of the detection system are both important; however, it should be noted that no amount of detector gain will compensate for a poor collection efficiency. Likewise, attempting to compensate for low efficiency or low detector gain with high gain

signal amplifiers will not make up for a poor detection system.

Other signal processing components may help to extract the weak signals from noise. These signal processors (such as baseline restorers and electronic filters) work on the assumption that noise is either random or highly correlated, whereas the desired signals are somewhere in between.

METHODS TO EVALUATE FLUORESCENCE SENSITIVITY

Since fluorescence sensitivity is dependent upon both collection (detection efficiency) and noise (background), it is not possible to define uniquely the operating condition of a flow cytometer with just one of these factors. This is demonstrated in Figure 1, with data obtained by adjusting the optical efficiency and background noise of an experimental flow cytometer to produce predetermined distributions for either the noise or a particular stained particle. Three different methods for assessing fluorescence sensitivity will be discussed in the context of this figure.

Delta Channel Method

The delta channel approach measures the difference between the mean or median channel of autofluorescent or nonfluorescent particles (or test pulse triggered background) and fluorescent particles. The separation between the peaks (delta channel value) is taken as a measure of

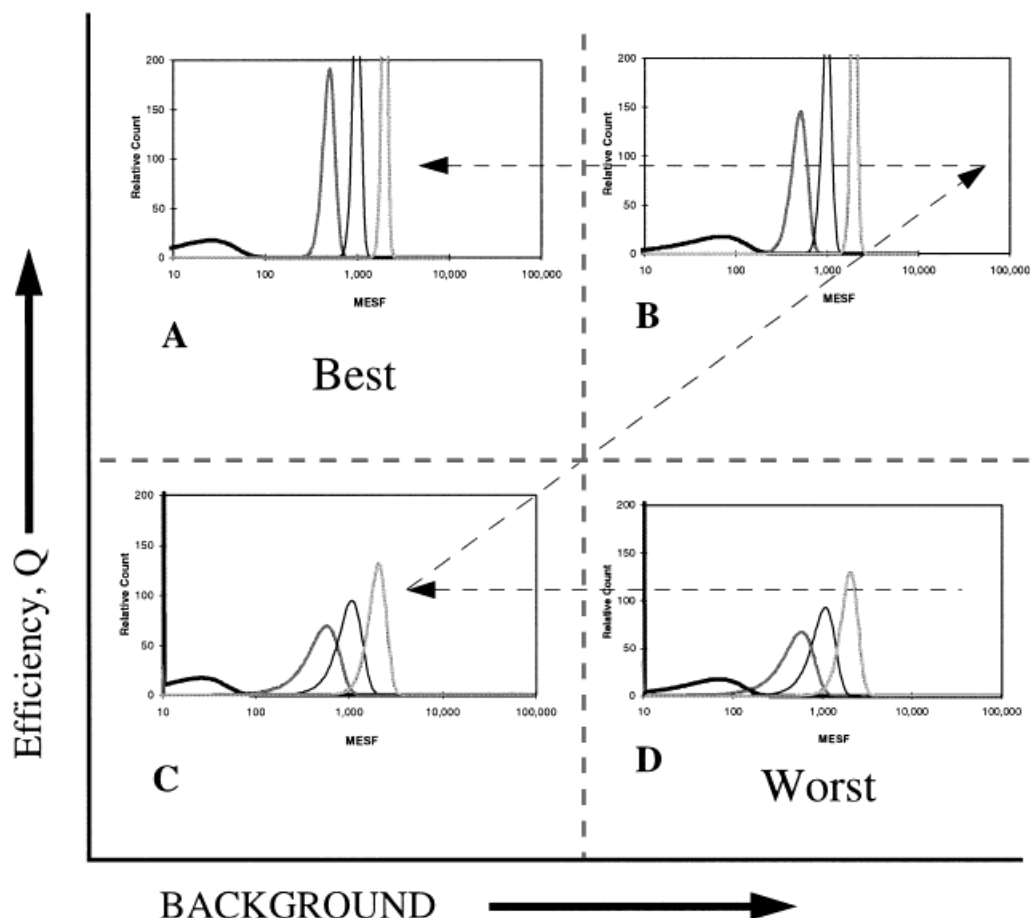


FIG. 2. Two-dimensional characterization of fluorescence sensitivity, which depends on both background noise and the efficiency (Q) with which the system converts fluorochromes to photoelectrons. Calculated histograms model an unstained particle and three stained particles. The stained particles in each histogram have the same amount of fluorochrome (MESF values of 500, 1,000, and 2,000), but background, Q , and the detector gain are varied to represent different instrument performance. Each population in this figure has the same number of events. The lowest peak (0 MESF) is the noise distribution. (The mean of the unstained particle is zero in each histogram; the apparent peak in the noise distribution is an artifact due to binning the normal distribution with a mean of zero into log channels.) In **B**, both Q and background are high. In **C**, both Q and background are low. Values were chosen to give the same

resolution of the noise distribution from the 500 MESF population. In **B** and **C**, 10% of events in the first "stained" population are below the noise threshold (defined as the channel number at which only 1% of noise events are above the channel). By this criterion, both **B** and **C** show the same resolution of the noise distribution from the dimmest stained population. **A** shows the same noise distribution as **C**, and **B** shows the same noise distribution as **D**. The resolution of the stained populations from the noise is significantly different when comparing **A** and **C** or **B** and **D**. Characterizing only one population or the overlap of the noise distribution and a distribution of stained particles does not uniquely characterize the ability of the instrument to resolve dimly stained populations.

sensitivity. This is illustrated with relatively bright particles in Figure 1A–E; and with dim particles in Figure 1K–O. For dim populations, the delta channel method does not critically assess the ability to resolve unstained from dimly stained particles, because the method takes into account only their mean or median channels and not the broadness of the fluorescence distributions.

Detection Threshold Method

This is an empirically derived method that characterizes the sensitivity of a flow cytometer by the apparent FI of an unstained particle (6). The implication is that particles which lie below this apparent FI cannot be detected or resolved.

The method requires a set of particles with a range of well-defined fluorescent intensities calibrated in terms of molecules of equivalent soluble fluorochrome (MESF). Linear regression is used to establish the best-fit line of histogram channel (usually the mean or median of the distribution) versus log MESF of the stained particle, and the line is extrapolated to the zero channel. The mean or median channel value of a blank (unstained) particle is converted to the corresponding MESF value on the extrapolated line, which is reported as the detection threshold in MESF units (6). Using the detection threshold method, conditions 2 and 3, and similarly conditions 4 and 5 in Figure 1 have the same apparent sensitivity.

This approach is widely used; it is used by most flow cytometry instrument manufacturers to set sensitivity

performance specifications. However, three considerations limit the value of the detection threshold. The most obvious limitation is presumed linearity required for the extrapolation, which extends into the least accurate range for analog logarithmic amplifiers. Moreover, the detection threshold can be misleading because of inaccuracies in locating the noise peak (Fig. 2) and its value is not only a function of the instrument's electronic and optical noise, but it is affected (and typically lowered) by the signal processing electronics such as baseline restorers. The different approaches to signal processing used by different cytometers makes direct comparison of detection threshold values questionable. Finally, the detection threshold is not truly the minimum detectable fluorescence signal, since any fluorescent particle will have a total fluorescence greater than a similar unstained particle. For example, a particle with a fluorescence intensity equal to the noise level will have a total fluorescence intensity of twice the noise level.

Qualitative Resolution Method

The most direct measure of resolution is to analyze a mixture of unstained and dimly-stained particles and observe the extent to which they are resolved (Fig. 1K–O). However, two instruments that are both able to resolve the same dim populations may do so to different degrees, and establishing quantitative units for this capability to resolve is problematic.

ESTABLISHING A MORE COMPLETE DEFINITION OF SENSITIVITY

Any single statistical measure of histogram distributions will fail to uniquely characterize the sensitivity of a cytometer. However, a two-dimensional characterization of sensitivity (Fig. 2) describes the full range of performance. If the background is low, the cytometer will show good resolution of a dimly stained population from noise, but unless the efficiency (Q) is high the population that is resolved well from noise will not be well resolved from a slightly brighter population. As a practical example consider a hypothetical case where lymphocyte autofluorescence is 5,000 MESF. We could have a situation where the

autofluorescence is well resolved from background noise, but where cells stained with 5,000 MESF (10,000 MESF total fluorescence) are not resolved from autofluorescence.

CONCLUSIONS

The characterization of a flow cytometer's capability to detect and to resolve dim fluorescent particles requires the knowledge of two instrument attributes namely background noise and detection efficiency. Current methods of defining are based on one or the other of these attributes. Because the methods are unidimensional, they do not provide a unique characterization or fail altogether to accurately predict the capability of a flow cytometer to detect and resolve dim fluorescent particles. We have shown the limitations of the current methodologies and propose that instrument sensitivity be defined in terms of background noise and detection efficiency. In order to uniquely determine the capability of a flow cytometer to detect and resolve dim fluorescent particles both attributes need to be considered together.

LITERATURE CITED

1. Brown MC, Hoffman RA, Kirchanski SJ: Controls for flow cytometers in hematology and cellular immunology. *Ann NY Acad Sci* 468:93–103, 1986.
2. Goodwin PM, Johnson ME, Martin JC, Ambrose WP, Marrone BL, Jett JH, Keller RA: Rapid sizing of individual fluorescently stained DNA fragments by flow cytometry. *Nucleic Acids Res* 21:803–806, 1993.
3. Hoffman RA: Standardization, calibration, and control in flow cytometry. In: *Current Protocols in Cytometry*, Unit 1.3, Robinson JP (ed). John Wiley & Sons, Inc., New York, 1997.
4. Horan PK, Muirhead KA, Slezak SE: Standards and controls in flow cytometry. In: *Flow Cytometry and Sorting*, Melamed MR, Lindmo T, Mendelsohn ML (eds). Wiley-Liss, New York 1990, pp 397–414.
5. Owens MA, Loken MR: *Flow Cytometry for Clinical Laboratory Practice: Quality Assurance for Quantitative Immunophenotyping*. Wiley-Liss, New York, 1995.
6. Schwartz A, Fernández-Repollet E, Vogt R, Gratama JW: Standardizing flow cytometry: Construction of a standardized fluorescence calibration plot using matching spectral calibrators. *Cytometry* 26:22–31, 1996.
7. Shapiro HM: *Practical Flow Cytometry*. Wiley-Liss, New York, 1995.
8. Steen HB: Noise, sensitivity, and resolution of flow cytometers. *Cytometry* 13:822–830, 1992.
9. Wood JCS: Clinical flow cytometry instrumentation. In: *Clinical Flow Cytometry: Principles and Application*, Bauer KD, Duque RE, Shankey TV (eds). Williams and Wilkins, Baltimore, MD, 1993, pp 71–92.